Auditable P2-YOLOv8 for Vulnerable Road Users: Multi-Seed Evidence and a Failed Degradation-Conditioned Localization Extension

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***Abstract*—Reliable road perception requires pedestrians and cyclists to remain localizable when distance, occlusion, and repeated downsampling leave weak spatial evidence. This study evaluates P2-YOLOv8, a YOLOv8m-derived detector with a shared anchor-free head at strides 4, 8, 16, and 32, and tests whether an exploratory degradation-conditioned localization learning (DCLI) extension adds value. DCLI combines supervised sampling and visibility factors with normalized distribution-focal-loss entropy while leaving feature fusion unchanged. The frozen comparison uses 3,341 fit images, a disjoint 371-image internal development subset derived from KITTI, five paired seeds, an epoch-30 endpoint, and Pedestrian/Cyclist Moderate AP_R40. PLAIN_P2 obtains** 95.3886 ± 0.5779 **macro AP_R40, compared with** 94.3702 ± 0.8473 **for DCLI. The paired PLAIN_P2-minus-DCLI mean is** $+1.0184$ **AP with a 95%** t **interval of** $[-0.4055, 2.4423]$ **and four positive pairs. Small- and far-object means also favor PLAIN_P2, although their intervals are wide and support is sparse. We therefore retain PLAIN_P2 as the tested development choice and reject a stable DCLI-gain claim. The result demonstrates that an active and numerically stable training mechanism is not evidence of improved terminal detection.**

***Index Terms*—vulnerable road users, pedestrian detection, cyclist detection, small-object detection, intelligent transportation, multi-seed evaluation**

I. Introduction

Pedestrians and cyclists are least protected when they are also hardest to observe: distant instances occupy few pixels, partial visibility removes shape cues, and a small localization error can move a box across a large fraction of the object. One-stage detectors must solve this problem while preserving the latency advantages that make them useful for road perception [1]. Their feature pyramids, however, commonly begin prediction at stride 8. At a 640-pixel input size, a 20-pixel-tall person spans only about 2.5 cells at that level. Deeper features add semantics, but they cannot recover fine boundaries already removed by downsampling.

Feature Pyramid Networks (FPN) established top-down semantic enrichment of high-resolution features [2], PANet added a bottom-up localization path [3], and EfficientDet studied efficient bidirectional fusion [4]. Small-object detectors similarly show that denser features can preserve useful evidence while increasing computation and background candidates [5], [6]. A stride-4 P2 head is therefore an established design family, including a recent direct pedestrian-detection precedent [7]. Our architectural question is deliberately narrower: can a frozen YOLOv8m-derived four-scale integration serve as a stable vulnerable-road-user detector under a fully specified internal protocol?

We additionally test whether the localization objective should react differently to two operational failure sources. Sampling degradation describes weak evidence caused by small support and resampling; visibility degradation describes missing support caused by occlusion or truncation. The exploratory DCLI mechanism predicts both factors, combines them with distributional-regression entropy, and applies a stopped-gradient scale to the complete-IoU (CIoU) residual [8], [9]. This construction is plausible but not self-validating. A finite gradient only shows that the path is active, and a favorable seed may reflect stochastic optimization rather than a stable method effect [10].

The study therefore separates three evidence layers. The first is architectural identity: PLAIN_P2 is the four-scale inference model. The second is mechanism identity: DCLI modifies the training loss while keeping assignment, DFL, and feature concatenation fixed. The third is terminal evidence: five paired seeds are evaluated only at epoch 30 with the same split, initialization rule, optimizer, prediction settings, and KITTI-style evaluator. The primary endpoint is the unweighted mean of Pedestrian and Cyclist Moderate AP_R40 on 371 internal development images. Scale/distance slices and a seed-0 cost audit are secondary diagnostics.

The results do not support the exploratory gain hypothesis. PLAIN_P2 has the higher five-seed mean, wins four of five paired macro comparisons, and is favored by the small and far diagnostic means. The macro interval nevertheless crosses zero, and the difficult slices have sparse support. We retain both facts: PLAIN_P2 is the defensible tested development choice, while population-level superiority is not established.

The paper makes three bounded contributions for reproducible road-perception development:

- an auditable stride-4-to-stride-32 P2-YOLOv8 integration and frozen evaluation protocol for vulnerable road users, without claiming the generic P2 principle as new;
- an explicit DCLI training hypothesis whose factors, entropy, stop-gradient semantics, calibration, and schedule are defined at implementation level; and
- a five-seed falsification record that separates mechanism activity, terminal AP, scale/distance diagnostics, and engineering cost, so the negative result narrows rather than inflates the method claim.

II. Related Work

a) High-resolution detection.: FPN and PANet make shallow spatial detail interact with deeper semantics through top-down and bottom-up paths [2], [3]; EfficientDet further studies weighted bidirectional fusion and scaling [4]. Scale Match frames tiny-person detection partly as a mismatch between training and target object scales [5], while QueryDet limits the cost of high-resolution prediction through sparse queries [6]. A recent workshop paper integrates a stride-4 P2 level into YOLOv12 for aerial pedestrians [7]. These works motivate high-resolution features but also delimit novelty: our contribution is a registered YOLOv8-derived road implementation and its evidence, not a new pyramid principle.

b) Dense assignment and localization.: Anchor-free FCOS predicts per-location box distances [11], and task-aligned methods such as ATSS and TOOD show that positive selection and classification–localization alignment materially affect dense detectors [12], [13]. We hold assignment fixed so that DCLI cannot inherit a gain from changed positives. Generalized Focal Loss represents box sides as discrete distributions and supplies DFL [8]. CIoU combines overlap, center distance, and aspect-ratio consistency [9]. DCLI uses normalized DFL entropy as one bounded input and transforms the CIoU residual, but does not change DFL or post-processing rank. VarifocalNet addresses confidence–localization alignment [14]; our score is training-only and is not a detection confidence.

c) Uncertainty and reproducible evidence.: Learned attenuation can reduce the influence of uncertain residuals, but such a scale is not automatically a calibrated posterior [15]. Recent small-object work also targets localization-uncertainty gradients through formulations distinct from DCLI [16]. In both cases, optimizer-side behavior must be judged by endpoint detection. Because initialization and stochastic training can change rankings, raw seed disclosure and paired comparisons are preferable to a selected run [10]. We use five pairs and a fixed endpoint; with only five pairs, intervals remain uncertainty summaries rather than definitive population tests.

III. P2-YOLOv8 and DCLI

A. Four-Scale P2 Detector

The stable detector, denoted PLAIN_P2, has a YOLOv8m-derived C2f/SPPF backbone [17], a bidirectional neck, and one shared anchor-free head over P2–P5 (Fig. 1a). Let B_2–B_5 be backbone tensors at strides 4, 8, 16, and 32. Starting from the SPPF-enhanced B_5 tensor, the neck constructs top-down tensors T_4 and T_3 and then P_2; a bottom-up path produces the final P_3–P_5 tensors:

$$\begin{aligned}
&T_4 = \mathrm{C2f}([\mathrm{Up}(B_5), B_4]), \quad T_3 = \mathrm{C2f}([\mathrm{Up}(T_4), B_3]),\\
&P_2 = \mathrm{C2f}([\mathrm{Up}(T_3), B_2]),\\
&P_3 = \mathrm{C2f}([\mathrm{Down}(P_2), T_3]), \quad P_4 = \mathrm{C2f}([\mathrm{Down}(P_3), T_4]),\\
&P_5 = \mathrm{C2f}([\mathrm{Down}(P_4), B_5]).
\end{aligned} \tag{1}$$

At 640×640, the four maps are 160^2, 80^2, 40^2, and 20^2. P2 contributes 25,600 of 34,000 spatial locations and approximately quadruples the cells covering an object relative to stride 8. This increases localization opportunity but can also increase ambiguous background candidates; resolution alone is not evidence of higher AP. Parameters are transferred by semantic prefix and compatible tensor shape. Newly introduced P2 and four-scale-head tensors are initialized by the frozen model definition, avoiding an invalid three-scale-head transfer.

The backbone outputs B_2–B_5 with 96, 192, 384, and 576 channels. The final Detect module consumes P_2, P_3, P_4, P_5 in that order, fixing the correspondence between feature level, stride, flattened prediction locations, DFL distributions, and the auxiliary factors introduced below. The order matters because a numerically compatible but semantically shifted factor field would condition the wrong anchors. The semantic-prefix transfer rule likewise separates inherited tensors from newly initialized P2 and four-scale-head tensors; it prevents an incompatible three-scale head from being counted as a common initialization.

PLAIN_P2 uses the standard weighted classification, CIoU, and DFL objective. Both arms retain the same backbone, neck, shared head, assignment, target-score normalization, DFL term, and evaluator. DCLI changes only the registered training pathway.

B. Degradation Factors and Uncertainty

DCLI estimates sampling and visibility factors $f_s, f_v \in [0, 1]$ at registered fusion nodes (Fig. 1b). Each estimator projects the two incoming features to 32 channels and concatenates them. A shared depthwise 3×3 convolution, GroupNorm, SiLU, 1×1 projection, GroupNorm, and SiLU precede a sigmoid two-channel head,

$$[f_s(x, y), f_v(x, y)] = \sigma(g_\theta(F_1, F_2)). \tag{2}$$

The factors have operational rather than latent-only semantics. Object-centred interventions provide reproducible targets: sampling applies a sub-pixel phase shift, area downsampling, and linear restoration; visibility mixes the region with a smooth low-frequency local-colour occluder. Identity, sampling, and visibility are drawn with probabilities 0.2, 0.4, and 0.4, and non-identity strength lies in $[0.1, 0.8]$. When a valid natural factor b and intervention strength a describe the same channel, the target is $t = 1 - (1 - b)(1 - a)$. The target is supported only on the transformed region and resized by area interpolation at each supervised scale. A validity-weight map makes missing natural metadata contribute zero weight rather than a false zero-degradation target.

The factor fields are flattened in the P2–P5 anchor order. Let $z_{a,q,k}$ be DFL logit k for anchor a and side $q \in \{l, t, r, b\}$, with $K = \texttt{reg\_max}$. The normalized mean entropy is

$$\begin{aligned}
p_{a,q,:} &= \mathrm{softmax}(z_{a,q,:}),\\
H_a &= -\frac{1}{4 \log K} \sum_q \sum_{k=1}^{K} p_{a,q,k} \log p_{a,q,k}.
\end{aligned} \tag{3}$$

Thus $H_a \in [0, 1]$ and is on the same bounded scale as the factors. The frozen score uses equal explicit weights,

$$u_a = \frac{f_{s,a} + f_{v,a} + H_a}{3}. \tag{4}$$

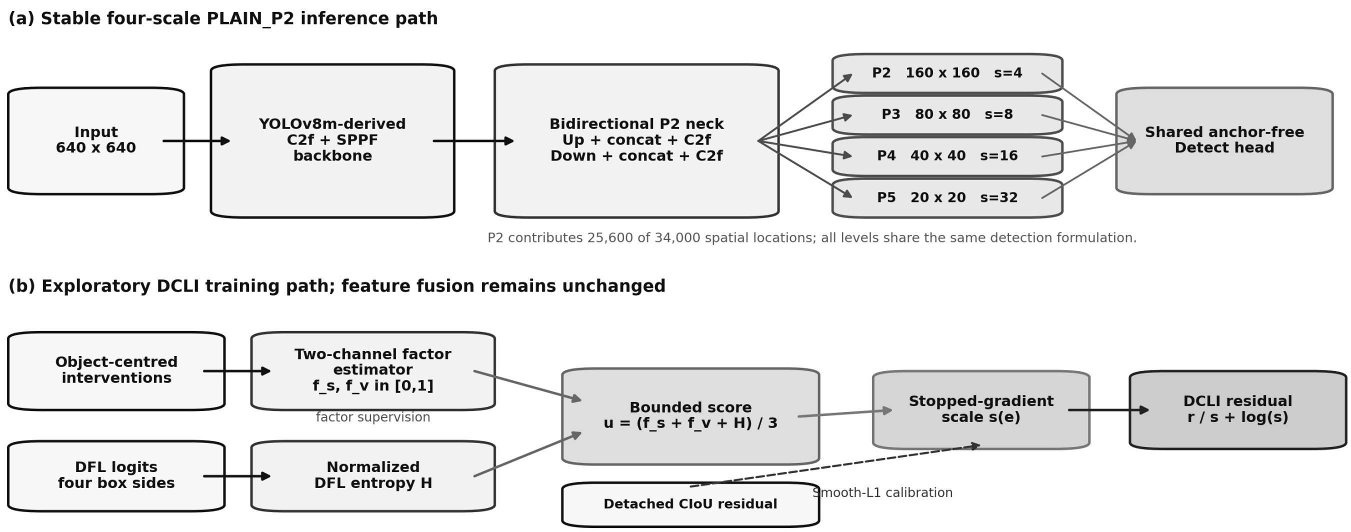

Fig. 1. Method overview. (a) PLAIN_P2 is the stable inference path: a YOLOv8m-derived backbone feeds a bidirectional neck and a shared head at strides 4–32. (b) DCLI is an exploratory training path. Learned degradation factors and normalized DFL entropy define a bounded score; stopped-gradient conditioning prevents a shortcut through the box loss. Feature concatenation remains unchanged in the frozen comparison.

This score is an optimization construct, not the probability that a detection is wrong. Separate validity-weighted multiscale Smooth-L1 supervision preserves the intended channel semantics. DFL itself, the assigned target, and the classification score remain unchanged.

C. Conditioned Localization and Boundary

For a positive anchor, let $r_a = 1 - \mathrm{CIoU}(b_a, b_a^*)$. CIoU geometry is evaluated in at least FP32. DCLI defines $s_a(e) = 1 + \beta\rho(e)\,\mathrm{stopgrad}(u_a)$ with $\beta = 0.5$ and replaces the CIoU residual by

$$\ell_{\mathrm{DCLI},a} = \frac{r_a}{s_a} + \log s_a. \tag{5}$$

The first term attenuates uncertain residuals; the logarithmic term resists an unnecessarily large scale. Stop-gradient is necessary: without it, the factor estimator could reduce the localization term by manipulating u_a instead of learning degradation semantics. If w_a is the registered target-score weight, both baseline and DCLI localization are normalized by the same $\sum_a w_a$. This freezes positive assignment and normalization rather than silently changing total localization scale.

A calibration term with gain $\gamma = 0.1$ matches u_a to the detached, clipped residual,

$$L_{\mathrm{cal}} = \gamma\rho(e)\frac{\sum_a w_a\,\mathrm{SmoothL1}(u_a, \mathrm{stopgrad}(\mathrm{clip}(r_a, 0, 1)))}{\sum_a w_a}. \tag{6}$$

Let L_{fac} be the validity-weighted multiscale factor loss and $\eta = 0.2$. The full exploratory objective is

$$L = \lambda_{\mathrm{cls}} L_{\mathrm{cls}} + \lambda_{\mathrm{box}}(L_{\mathrm{DCLI}} + L_{\mathrm{cal}}) + \lambda_{\mathrm{dfl}} L_{\mathrm{DFL}} + \eta\rho(e) L_{\mathrm{fac}}. \tag{7}$$

The absolute zero-based epoch defines a resume-stable schedule,

$$\rho(e) = \begin{cases} 0, & e < 5, \\ \min\{1, (e - 5 + 1)/10\}, & e \geq 5. \end{cases} \tag{8}$$

It is zero for epochs 0–4, increases from 0.1 to 1.0 over epochs 5–14, and remains one thereafter. Input interventions stay enabled independently; the schedule controls the loss pathways rather than rewriting the data policy.

The frozen boundary is essential. Feature-fusion gating, semantic protection, and counterfactual consistency are disabled; effective fusion-gate and counterfactual gains are zero. DCLI does not alter feature concatenation at inference, although the registered auxiliary estimator is still instantiated and its measured cost is reported. Gradients from conditioning, calibration, and factor supervision can verify pathway activity and finite optimization, but they cannot replace terminal AP. The experiment tests the complete DCLI bundle—interventions, factor supervision, calibration, and conditioning—not a factorial attribution of its components.

Table I prevents two common attribution errors. First, the comparison cannot be described as feature gating because the fusion path is ordinary concatenation in both arms. Second, it cannot isolate the factor estimator from its intervention, calibration, and conditioned-loss dependencies. The estimand is the registered DCLI bundle on a shared P2 detector.

IV. Experimental Protocol

A. Data, Endpoint, and Integrity

Experiments use KITTI labelled object-detection images and annotations [18]. The project first freezes a disjoint 3,712/3,769 partition of the 7,481 labelled images. The comparison is confined to the 3,712-image side: 3,341 identifiers form the fit subset and 371 different identifiers form the

TABLE I

Frozen method boundary. "Same" denotes an identical registered component in both arms; disabled components have zero effective gain.

Component	PLAIN_P2	DCLI	Role in the comparison
P2–P5 detector, assignment, DFL	Same	Same	Holds inference topology, positives, and distributional regression fixed.
Object-centred interventions	Off	On	Supplies operational sampling/visibility targets; part of the tested bundle.
Two-channel factor estimator	Off	On	Predicts bounded sampling and visibility fields at registered fusion nodes.
Conditioned CIoU / calibration	Off	On	Changes the per-positive localization objective with frozen gains.
Feature-fusion gate	Off	Off	Factors never reweight or route inference features.
Semantic protection	Off	Off	Excludes an additional semantic adapter from the comparison.
Counterfactual consistency	Off	Off	Gain is zero; no consistency claim is attached to this experiment.

internal development subset; their disjoint union reproduces that parent partition. The separate 3,769-image side is outside the comparison. The detector trains on Car, Pedestrian, and Cyclist, but the primary endpoint averages only Pedestrian and Cyclist Moderate AP_R40. The 371 images are neither the official KITTI validation nor the official test set.

A read-only audit found all 3,712 images decodable as RGB, no exact image-content hash duplication within or across fit/development, and no malformed, non-finite, duplicate, or non-positive-area main-class annotation row. Raw labels contain 15,460 main-class objects in fit and 1,838 in development (17,298 in the 3,712-image parent); the processed-label row count agrees. These checks establish identifier, file, and exact-content integrity, not the absence of broader scene similarity or development-set adaptation.

The frozen Moderate filter requires raw box height $y_2 - y_1 > 25$ pixels, occlusion ≤ 1, and truncation ≤ 0.30. The project evaluator handles *Person_sitting* and *DontCare* through its registered ignore logic. With interpolated precision $p(c, r)$, class AP is

$$\mathrm{AP\_R40}(c) = \frac{100}{40} \sum_{j=1}^{40} p(c, j/40), \qquad m = \tfrac{1}{2}(\mathrm{AP\_R40}_{\mathrm{Ped}} + \mathrm{AP\_R40}_{\mathrm{Cyc}}). \tag{9}$$

Calling this metric "KITTI-style" records that the rules are applied locally to the internal development subset, not submitted to the KITTI server.

B. Matched Training and Evaluation

Both arms use the same P2 model definition, 640-pixel input, batch 16, eight workers, 30 epochs, data split, augmentation policy, initialization checkpoint/transfer rule, prediction settings, and evaluator. SGD uses initial learning rate 0.01, final factor 0.01, momentum 0.937, weight decay 0.0005, and a three-epoch warm-up. AMP and deterministic execution are enabled; caching is disabled. Five registered seeds (0–4) control matched data and model stochasticity; DCLI-specific parameters use the corresponding registered base seed. The formal checkpoint is `last.pt` at epoch 30. No early stopping or best-epoch selection is permitted. Surviving metadata records configured batch 16 and nominal accumulation four, but not the actual accumulation state; the paper therefore does not promote nominal batch 64 to an observed effective batch.

The shared upstream augmentation policy is inherited from the frozen runtime rather than rewritten as a contemporary-library default. DCLI-only object interventions are therefore recorded as mechanism inputs, not mislabelled as matched baseline augmentation. Pairing also includes the semantic-prefix initialization rule: every arm starts from the same compatible inherited tensors, while new or DCLI-specific tensors are controlled by the registered seed.

Prediction uses confidence 0.001, NMS IoU 0.70, at most 300 detections per image, and FP32 output. Pedestrian and Cyclist matching uses IoU 0.5. For seed i, m_i is the mean of the two class AP_R40 values, and $d_i = m_i^{\mathrm{PLAIN}} - m_i^{\mathrm{DCLI}}$. For $n = 5$,

$$\bar{d} = \frac{1}{n} \sum_i d_i, \quad \mathrm{CI}_{95} = \bar{d} \pm t_{0.975,4} \frac{\mathrm{SD}(d)}{\sqrt{n}}. \tag{10}$$

We report method means, sample SDs, every d_i, the interval, and sign counts. With four degrees of freedom, the distributional assumption cannot be stress-tested; the interval is an uncertainty summary, not a distribution-free test. An interval containing zero is insufficient evidence of stable directional superiority, not proof of equality. No multiple-comparison correction or confirmatory claim is attached to class and slice diagnostics.

The pre-frozen DCLI claim required a macro gain of at least +1.1 AP in the DCLI-favouring direction, not merely a positive selected seed. We preserve that decision rule separately from null-hypothesis language: failure of the gain gate selects neither significance nor equivalence. It simply prevents the registered exploratory mechanism from being promoted as the development detector. This distinction is important because a noisy estimate can miss a practical gain gate while its interval also fails to establish a nonzero population effect.

The small slice is $25 <$ raw KITTI box height ≤ 40 pixels after Moderate filtering; far is depth > 40 m; near is $0 <$ depth

TABLE II

Frozen experimental contract. Values are shared unless explicitly marked DCLI-only.

Dimension	Frozen value	Dimension	Frozen value
Fit / development	3,341 / 371 images	Seeds	0, 1, 2, 3, 4
Architecture / input	P2-YOLOv8m / 640^2	Epoch endpoint	30, `last.pt` only
Batch / workers	16 / 8	Optimizer	SGD, lr 0.01, momentum 0.937
Final LR factor / decay	0.01 / 0.0005	Warm-up	3 epochs
AMP / deterministic / cache	on / on / off	Runtime identity	Ultralytics 8.4.98; Python 3.12
Prediction	conf 0.001; NMS 0.70; max 300	Output / matching	FP32; Ped./Cyc. IoU 0.5
DCLI-only schedule	5 frozen + 10 ramp epochs	Formal exclusions	best epoch; 15-epoch/batch-2 screens

$\leq$ 20 m; and large is raw height > 80 pixels. These are diagnostics: small/far support is only 29/6 Pedestrian/Cyclist and 9/2, respectively. Training curves are optimization diagnostics and do not share the KITTI AP_R40 semantics. Engineering cost is measured for the matched seed-0 pair only (batch 1, 640 pixels, FP32, 50 warm-ups, five synchronized passes over 371 images).

C. Reproducibility and Evidence Identity

The registered model YAML, initialization checkpoint, method specification, terminal table, paired-statistics file, and canonical-run manifest are bound by SHA-256 in the evidence ledger. A seed-normalized audit verifies all ten formal configurations, including within-method seed-only differences and equality of shared training and prediction sections for each pair. The runtime contract pins Ultralytics 8.4.98 and a Python 3.12 executable. Interrupted/nonterminal runs and every 15-epoch, batch-2 local direction screen are excluded from the formal evidence. The receipts do not retain a complete Torch/CUDA/driver lock for every run, so we bind implementation identity to frozen source/package artifacts rather than claim an unobserved environment state.

V. Results and Discussion

A. Five-Seed Terminal Evidence

PLAIN_P2 obtains 95.3886 ± 0.5779 macro AP_R40, versus 94.3702 ± 0.8473 for DCLI (Table III). The five paired differences are −0.3423, +2.2081, +2.0475, +1.1212, and +0.0574 AP. DCLI wins only seed 0; PLAIN_P2 wins seeds 1–4. A seed-0-only report would therefore reverse the five-seed development decision.

The macro paired mean is +1.0184 AP, with sample SD 1.1468 and 95% t interval $[-0.4055, 2.4423]$. The interval includes zero. Pedestrian has the larger mean difference (+1.5987, interval $[-0.3727, 3.5701]$, four positive pairs); Cyclist has +0.4381 ($[-0.9337, 1.8099]$, three positive pairs). The data support selecting PLAIN_P2 within this finite development comparison, but they do not establish population-level significance or universal superiority.

Figure 2 makes the seed dependence and interval width explicit. The outcome also fails the pre-frozen DCLI gain requirement of at least +1.1 AP in the DCLI-favouring direction. The observed direction is instead +1.0184 for PLAIN_P2. We retain the failure rather than tune the claim after seeing the result.

B. Scale/Distance Diagnostics and Cost

The difficult slices do not reveal a hidden DCLI benefit (Table IVa). For small objects, PLAIN_P2 reaches 67.5344 macro AP_R40 and DCLI 56.3170; the paired mean is +11.2175 for PLAIN_P2 with interval $[-1.3942, 23.8291]$ and four positive pairs. For far objects, the corresponding means are 57.7167 and 47.6637; the difference is +10.0530 with interval $[-9.0897, 29.1957]$ and four positive pairs. The intervals are wide, and the two far-Cyclist objects make that class especially discrete. Near objects favor PLAIN_P2 by +0.8375 with five positive pairs, while large objects show a small +0.4483 mean with mixed signs. No confirmatory no-harm margin was frozen for near/large, so they remain descriptive.

Both arms use the same P2 architecture; these differences estimate the DCLI bundle on P2, not a P2-versus-no-P2 effect. The missing matched three-scale control is a central limit on the architectural claim.

DCLI also incurs local measured overhead (Table IVb). It adds 0.482% parameters and 1.281% FLOPs; in the seed-0 audit, median latency rises 18.305%, throughput falls 15.428%, peak VRAM rises 9.906%, and training runtime rises 26.980%. The larger wall-clock deltas can include framework and measurement effects and are not a device population. They nevertheless do not rescue an accuracy–cost trade-off: the registered extension has lower mean AP and additional local cost.

C. Trajectory Diagnostics and Endpoint Discipline

The surviving seed-0/1 curves also enforce endpoint discipline. From epochs 20 to 30, Ultralytics validation mAP50–95 changes from 0.65274 to 0.68060 and 0.63900 to 0.68886 for PLAIN_P2, and from 0.62876 to 0.68433 and 0.64287 to 0.66991 for DCLI. Every run still moves after epoch 20, while the method ranking reverses across seeds; selecting an intermediate or best checkpoint could therefore change the conclusion. These curves are not pooled with the primary endpoint because their IoU range and evaluator differ from Moderate Pedestrian/Cyclist AP_R40. Likewise, DCLI changes the box objective, so its loss magnitude is not directly comparable with baseline CIoU. Finite curves establish completed optimization; only terminal AP supports the ranking.

D. Interpreting the Negative Result

The frozen pathway is active and numerically stable, yet the endpoint deteriorates on average. Three explanations are plausible but not proven. First, attenuation may suppress

TABLE III

Epoch-30 Moderate AP_R40. Differences are PLAIN_P2 minus DCLI and are computed before display rounding.

(a) Pedestrian/Cyclist macro AP_R40 by seed

Seed	PLAIN_P2	DCLI	Difference
0	95.2506	95.5928	−0.3423
1	95.7872	93.5790	+2.2081
2	95.6245	93.5770	+2.0475
3	95.8401	94.7189	+1.1212
4	94.4408	94.3835	+0.0574
Mean	95.3886	94.3702	+1.0184
Sample SD	0.5779	0.8473	1.1468

(b) Paired endpoint summaries

Endpoint	Mean diff.	95% t interval	Wins
Pedestrian	+1.5987	$[-0.3727, 3.5701]$	4/5
Cyclist	+0.4381	$[-0.9337, 1.8099]$	3/5
Macro	+1.0184	$[-0.4055, 2.4423]$	4/5

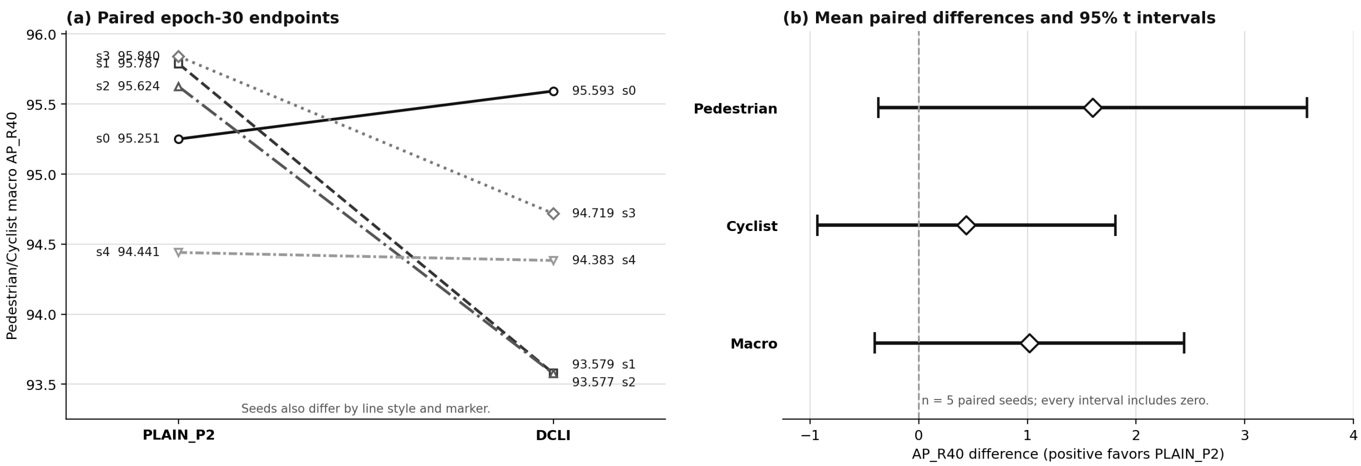

Fig. 2. Five-seed terminal evidence. (a) Each line links the two methods for one registered seed. (b) Points are paired means and bars are two-sided 95% t intervals. All intervals include zero; positive differences favor PLAIN_P2.

TABLE IV

Diagnostic slices and matched seed-0 cost. Slice intervals are descriptive; cost is one local matched pair.

(a) Five-seed diagnostic macro AP_R40

Slice	PLAIN	DCLI	Mean diff.	95% t interval	Wins
Small	67.5344	56.3170	+11.2175	$[-1.3942, 23.8291]$	4/5
Far	57.7167	47.6637	+10.0530	$[-9.0897, 29.1957]$	4/5
Near	96.7938	95.9563	+0.8375	$[0.1414, 1.5336]$	5/5
Large	97.1207	96.6724	+0.4483	$[-0.6989, 1.5956]$	2/5

(b) Seed-0 engineering measurements

Metric	PLAIN	DCLI	Rel. change
Parameters (M)	25.053	25.173	+0.482%
FLOPs (G)	97.932	99.187	+1.281%
Latency (ms)	20.595	24.365	+18.305%
FPS	48.505	41.022	−15.428%
Peak VRAM (MB)	900.184	989.359	+9.906%
Train time (s)	1048.03	1330.79	+26.980%

the boundary corrections most needed for small or distant instances. The log-scale term prevents unlimited attenuation, but it cannot ensure that gradient mass is reallocated in a task-beneficial way. This explanation is consistent with the lower DCLI small/far means; it is not a causal attribution because no residual-by-scale intervention is available.

Second, the factor targets are operational proxies. Synthetic area resampling and smooth occlusion do not reproduce the full joint distribution of motion blur, sensor noise, truncation, texture loss, and annotation ambiguity in road scenes. A factor estimator can therefore fit its supervision while the resulting uncertainty ordering is poorly aligned with the box corrections that improve AP. Third, the high absolute internal-development scores leave limited headroom, while the auxiliary estimator, calibration, and ramp add seed-sensitive optimization degrees of freedom. The larger DCLI sample SD and the seed-0 ranking reversal are compatible with this account, but five pairs cannot identify which component is responsible.

The result establishes three non-equivalences. A finite mechanism gradient is not an AP gain. One favorable seed is not a stable method effect. A general validation mAP50–95 curve is not the KITTI-style Pedestrian/Cyclist Moderate AP_R40 endpoint. These distinctions are the diagnostic contribution of the negative result. They justify the practical decision to retain PLAIN_P2 and prevent DCLI from being presented as a validated improvement. They do not imply that all uncertainty-aware localization is ineffective; the falsified object is this frozen bundle, schedule, and data protocol.

This negative result changes the next experiment. A productive revision should not tune β, γ, or the ramp against the same five terminal outcomes. It should first test the highest-ranked mechanism explanation with a preregistered, single-variable comparison—for example, whether attenuation is disproportionately assigned to small/far positives—and preserve the same endpoint and seed pairing. Separately, a conventional three-scale control is required before any causal P2 gain can be estimated. These are distinct questions: one diagnoses why DCLI failed on a P2 detector; the other measures what the P2 branch contributes relative to its matched architectural parent.

E. From Evidence to a Development Decision

The retained model decision and the scientific claim answer different questions. PLAIN_P2 has the higher observed mean, lower sample SD, four paired wins, favorable difficult-slice means, and lower seed-0 cost; DCLI also fails its preregistered gain gate. These observations make PLAIN_P2 the lower-risk development choice between the tested arms. Scientific inference is stricter: the primary and class-wise intervals include zero, small/far support is sparse, and no equivalence margin was frozen. The conclusion is therefore local—DCLI does not establish the required gain under this split, endpoint, horizon, and seed set; neither superiority nor equivalence is established.

The architectural claim is narrower still. Both arms contain P2, so neither the macro difference nor the small-object slice estimates a causal P2 effect. A matched three-scale control is required. The negative mechanism result remains useful because each artifact exposes one failure mode: seed selection, sparse slices, intermediate-checkpoint selection, or hidden cost. None is allowed to support a stronger claim than it measures.

F. Limitations and Threats to Validity

The 371-image development subset is not an untouched hidden test set, and repeated project decisions may adapt to it. Hashes rule out exact duplication, not correlated frames or shared scenes. Five seeds leave wide intervals; the 30-epoch horizon does not prove asymptotic convergence. The endpoint excludes Car and Easy/Hard conditions, and no external dataset tests adverse weather, cameras, or domains.

No matched three-scale control identifies a P2 effect. The DCLI arm tests a bundle, not isolated intervention, supervision, calibration, or conditioning effects. Cost is one seed and one local environment. The runtime identity fixes implementation and Ultralytics 8.4.98, but not every Torch/CUDA/driver or actual accumulation state. PLAIN_P2 is therefore the retained tested choice, not a universally optimal detector.

External validation should preserve the same separation of claims. An untouched road dataset can test transfer of the retained four-scale detector, whereas a preregistered three-scale arm can identify the architectural effect. A revised DCLI mechanism should be evaluated only after its single changed assumption and gain criterion are frozen. Combining these steps would again make architecture, mechanism, and domain shift inseparable.

VI. Conclusion

Across five paired seeds, PLAIN_P2 attains 95.3886 ± 0.5779 macro Moderate AP_R40 and DCLI 94.3702 ± 0.8473; the paired +1.0184 AP mean favors PLAIN_P2, but its interval includes zero. Difficult slices and seed-0 cost reveal no countervailing DCLI benefit. We retain PLAIN_P2 and report DCLI as a falsified exploratory extension, not an improvement. A matched three-scale control, untouched external data, and preregistered mechanism revision are the next tests.

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